A Two-Wave, Pilot-Selected Stratification Design with Precision-Weighted Combination

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This memo describes the two-stage ("two-wave") estimator implemented in `one_iter_pilot_select()` in the `stratsampling` package: a small pilot is used to *choose* a stratification, a second, larger wave is drawn under that stratification, and the two waves' treatment-effect estimates are then combined with a precision (inverse-variance) weighted average. The companion memo `scripts/pilot_shrinkage_memo.md` covers how the per-stratum means and variances used along the way are themselves empirical-Bayes shrunk; this memo focuses on the two-wave design and, in particular, derives the weighting formula used to combine the two waves' estimates.

1 Setup and Notation

We have a population of N units with treatment indicator $Z_i \in \{0, 1\}$, an eventual human-coded outcome Y_i, and a surrogate score $\hat{Y}_i$ available for every unit. A total coding budget of n units can be human-coded. We have a library of candidate stratifications $\{S^{(1)}, \ldots, S^{(M)}\}$, each a partition of the population into strata, built from covariates and $\hat{Y}_i$ before any coding occurs. The target of inference is the average treatment effect τ.

The design splits the budget n into a pilot wave of size n_1 (a fixed fraction $\rho \in (0, 1)$ of n) and a main wave of size $n_2 = n - n_1$.

2 Stage 1: Pilot Sample and Stratification Selection

A simple random sample (within treatment arm) of size n_1 is coded; call this pilot set P. For each candidate stratification $S^{(m)}$, the pilot's coded residuals $e_i = Y_i - \hat{Y}_i,\ i \in P$, are used to estimate the (shrinkage-adjusted) between-stratum spread of mean residuals net of within-stratum noise,

$$\hat{D}_m \;=\; \widehat{BS}_m \;-\; \widehat{WS}_m,$$

where $\widehat{BS}_m$ estimates how much the stratum-mean residuals of $S^{(m)}$ differ from each other (the signal a stratification needs to be useful) and $\widehat{WS}_m$ estimates the average residual noise remaining within a stratum (see `compare_stratifications_precoding()` and the shrinkage

memo for how $\widehat{BS}_m$ and $\widehat{WS}_m$ are computed from the pilot). The selected stratification is

$$S^* \;=\; \underset{m\in\{1,\dots,M\}}{\arg\max}\; \hat{D}_m.$$

The pilot also yields a first estimate of the treatment effect, the (unbiased) model-assisted SRS estimator computed on P alone,

$$\hat{\tau}_1, \qquad \hat{V}_1 \;=\; \widehat{\mathrm{Var}}(\hat{\tau}_1).$$

3 Stage 2: Main Sample on the Selected Stratification

A fresh sample of size n_2 is then drawn from the *full* population — not the pilot's uncoded remainder — stratified on S^*, with allocation either proportional to stratum size or Neyman-optimal using the pilot's (shrunk) variance estimates as plug-in values. This yields the wave-2 model-assisted stratified estimator

$$\hat{\tau}_2, \qquad \hat{V}_2 \;=\; \widehat{\mathrm{Var}}(\hat{\tau}_2),$$

again unbiased for τ. Drawing wave 2 from the full population, rather than only the unpiloted remainder, keeps $\hat{\tau}_2$'s design close to a standard stratified sample on S^*; the tradeoff, and the (approximate) independence of $\hat{\tau}_1$ and $\hat{\tau}_2$ that this buys, is discussed in Section 5.

4 Combining the Two Waves: Precision Weighting

We now have two unbiased estimators of the same τ, $\hat{\tau}_1$ and $\hat{\tau}_2$, treated as independent (Section 5), with variances V_1, V_2. We then average them with a weighted average

$$\hat{\tau}(w) \;=\; w\,\hat{\tau}_1 + (1-w)\,\hat{\tau}_2, \qquad w \in [0,1].$$

where w is based on the (estimated) precisions of the two estimates:

$$w^* \;=\; \frac{V_2}{V_1+V_2} \;=\; \frac{1/V_1}{1/V_1+1/V_2},$$

and at this weight the combined estimator's variance satisfies

$$\mathrm{Var}(\hat{\tau}(w^*)) \;=\; \frac{V_1V_2}{V_1+V_2}, \qquad \text{equivalently} \qquad \frac{1}{\mathrm{Var}(\hat{\tau}(w^*))} \;=\; \frac{1}{V_1}+\frac{1}{V_2}.$$

The last identity is the familiar "precisions add" result: combining two independent unbiased estimates by inverse-variance weighting can only help, and never hurts more than either wave alone, since $1/V_1+1/V_2 \geq \max(1/V_1, 1/V_2)$. This is the same weighting principle behind the fixed-effect model in meta-analysis [DerSimonian and Laird, 1986] and classical combination of independent survey estimates [Cochran, 1977], and, in a different guise (borrowing strength across many strata instead of across two waves), behind the empirical-Bayes

shrinkage in the companion memo [Fay and Herriot, 1979]. (These are claude references and might be fiction–Check)

In practice V_1, V_2 are unknown and are replaced by their estimates $\hat{V}_1 = \widehat{SE}_1^2$, $\hat{V}_2 = \widehat{SE}_2^2$, giving the feasible combination implemented by `combine_estimators()`:

$$\hat{w} \;=\; \frac{\hat{V}_2}{\hat{V}_1 + \hat{V}_2}, \qquad \hat{\tau}_{\text{combo}} \;=\; \hat{w}\,\hat{\tau}_1 + (1-\hat{w})\,\hat{\tau}_2, \qquad \widehat{SE}(\hat{\tau}_{\text{combo}}) \;=\; \sqrt{\hat{w}^2\hat{V}_1 + (1-\hat{w})^2\hat{V}_2}.$$

Note: Each wave's estimator variance is itself sometimes reported as a sum of sub-components (e.g. a treatment/allocation component plus a sampling component); `combine_estimators()` applies the *same* $\hat{w}$ to each sub-component of each wave, if there is interest in those sub-components. This means combining each final sub-component separately, with the one weight w^*, adds back up to the combination of the whole.

5 Assumptions and Caveats

Two assumptions are doing real work above and are worth stating plainly.

Unbiasedness of each wave. $\hat{\tau}_1$ is an SRS-based model-assisted estimator computed on the pilot alone, and $\hat{\tau}_2$ is a stratified model-assisted estimator computed on a fresh wave-2 sample; each is unbiased for τ on its own design, *provided* the choice of S^* is not itself informed by wave 2's realized data. Selecting S^* using only pilot residual spread, before wave 2 is drawn, keeps that choice a (random, pilot-dependent) but fixed design feature from wave 2's point of view.

Independence of the two waves. The precision-weighting formula in Section 4 assumes $\hat{\tau}_1 \perp\!\!\!\perp \hat{\tau}_2$. Wave 2 is drawn from the *full* population rather than the un-piloted remainder specifically so that its design does not structurally depend on which units the pilot happened to code. But because wave 2 is drawn from the full population, it can by chance recode a unit that was already coded in the pilot; that unit's residual then contributes to both $\hat{\tau}_1$ and $\hat{\tau}_2$, inducing a small positive correlation between them that the formulas for the standard errors ignore. The pilot fraction ρ is small by design, so this overlap, and the correlation it induces, should be a second-order effect in most regimes; treating it as exactly zero is a simplification, not a proven result, and the combined $\widehat{SE}(\hat{\tau}_{\text{combo}})$ above should be read as mildly anti-conservative to the extent that overlap occurs.

References

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Robert E. Fay and Roger A. Herriot. Estimates of income for small places: An application of james-stein procedures to census data. *Journal of the American Statistical Association*, 74(366):269–277, 1979.